

Keith L. Baskerville Jr.

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Backend / Applied AI Engineer — Python • FastAPI • Django • PostgreSQL • AI/LLM Systems

Backend and applied AI engineer. I built and shipped Symbiosis solo — semantic search, ETL, and LLM extraction over 5,900+ papers — and spent four years before that teaching backend engineering to 300+ students. That combination is the point: I can build the retrieval system and then sit with the people who have to use it.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

Backend: FastAPI, Django, Django REST Framework, Node.js, Express — REST API design, JWT auth, MVC, system design, data modeling

Data: PostgreSQL, Elasticsearch, Redis, MongoDB, DynamoDB, vector databases — ETL pipelines, indexing, caching

AI/ML: RAG, embeddings, hybrid search, structured extraction, vision-model prompting, LangGraph, FastMCP, agent workflows

Infra: Google Cloud (Cloud Run, Cloud Build), Vercel, Elastic Cloud, AWS, Docker, CI/CD, Git, Railway

Frontend: React, Next.js

EXPERIENCE

Full Stack Engineer — Equitable AI Underwriters Lab | Remote | Oct 2025 – Apr 2026

Research lab applying systems thinking and ML to equitable societal outcomes. Symbiosis was originally developed at Google as a closed-source project; I rebuilt and extended it with its lead researcher.

- Sole engineer on Symbiosis, a production semantic search platform over 5,900+ system dynamics papers and 1,200+ machine-extracted causal models: FastAPI, Next.js, Elasticsearch, 3,072-dimensional embeddings, shipped to Cloud Run & Vercel.
- Built the ETL pipeline over 3,286 PDFs & 72k+ extracted figures, producing 884 structured causal diagrams at an 89% parse rate.
- Converted 327 stock-and-flow models into queryable graphs via a custom XMILE/Vensim parser, and recovered orphaned model-to-paper links with an embedding-similarity auto-linker.
- Cut retrieval latency with hybrid BM25 + vector search over precomputed embeddings, collapsing per-paper N+1 lookups into one batched query and dropping timeouts from 60s to 5s.
- Owned the Elasticsearch schema, OAuth, and Cloud Build CI/CD; a single gateway proxy made the move to Elastic Cloud a config change instead of a frontend rewrite.

Technical Instructor & Curriculum Facilitator | General Assembly • The Last Mile • Codepath | 2022 – Present

- Taught backend and full stack engineering — Python, JavaScript, React, REST APIs, MVC — to 300+ students across three programs, including in-person instruction for incarcerated students at The Last Mile.
- Led code reviews, debugging sessions, and technical assessments; 85% course completion across 7 cohorts at General Assembly.
- Prepared 60+ Codepath fellows for engineering internships, with fellows placing at companies including Salesforce.

Earlier Career: Glide Memorial • Compass Family Services | San Francisco, CA | 2017 – 2020

Ran financial-literacy and housing-access programs; co-designed an internal housing database with the Data Impact team.

PROJECTS

Rights2Roof — Multi-Agent AI Housing Rights Assistant | Runner-up out of 7 teams, Google x Kura Labs Agentic AI competition

github.com/OhziiLov3/rights2roof · vimeo.com/1143307938

- Built a multi-agent tenant-rights assistant with LangGraph, FastMCP, FastAPI, and Slack — a 3-node planner → RAG → executor graph exposing 9 MCP tools over 333 pages of California and New York housing law.
- Implemented Redis-backed vector search (text-embedding-3-large, top-5 retrieval) alongside conversation state, caching, and per-user rate limiting across 4 decoupled Railway services, cutting redundant LLM and API calls on follow-up turns.

Spark Properties — Full-Stack Property Rental Marketplace | spark-properties.vercel.app · github.com/OhziiLov3/Spark-Properties

- Built and shipped a property rental marketplace solo — Next.js 15 server actions, MongoDB/Mongoose, and Google OAuth with NextAuth, with route protection and per-user listing management.
- Implemented Mapbox geocoding and location search, Cloudinary multi-image upload and optimization, paginated listing queries, and an internal user-to-user messaging system with unread notifications.

EDUCATION & CERTIFICATIONS

University of San Francisco — Master of Arts, International Studies, 2025

Thesis: Financial Inclusion — The Role of Banks and Credit Unions in San Francisco's Civic Center–Tenderloin (mixed-methods research).

Stockton University, NJ — Bachelor of Arts, Philosophy and Religion; Minor in Economics, 2015

AWS Certified Cloud Practitioner, 2025 • General Assembly Software Engineering Immersive, 2021